

Cheat Sheet: Plotting with Matplotlib

1 Plotting with Matplotlib Using Pandas

Plot Type	Purpose	Pandas Function	Example
Line Plot	Shows trends and changes over time	<code>DataFrame.plot.line()</code> <code>DataFrame.plot(kind='line')</code>	<code>df.plot(x='year', y='sales', kind='line')</code>
Area Plot	Displays data series as filled areas, showing the relationship between them	<code>DataFrame.plot.area()</code> <code>DataFrame.plot(kind='area')</code>	<code>df.plot(kind='area')</code>
Histogram	Displays bars representing the data count in each interval/bin	<code>Series.plot.hist()</code> <code>Series.plot(kind='hist', bins=n)</code>	<code>s.plot(kind='hist', bins=10)</code> <code>df['age'].plot(kind='hist', bins=10)</code>
Bar Chart	Displays data using rectangular bars	<code>DataFrame.plot.bar()</code> <code>DataFrame.plot(kind='bar')</code>	<code>df.plot(kind='bar')</code>
Pie Chart	Displays data as a circular plot divided into slices, representing proportions or percentages of a whole	<code>Series.plot.pie()</code> <code>Series.plot(kind='pie')</code> <code>DataFrame.plot.pie(y, labels)</code> <code>DataFrame.plot(kind='pie')</code>	<code>s.plot(kind='pie', autopct='%1.1f%%')</code> <code>df.plot(x='Category', y='Percentage', kind='pie')</code>
Box Plot	Displays the	<code>DataFrame.plot.box()</code> <code>DataFrame.plot(kind='box')</code>	<code>df_can.plot(kind='box')</code>

	distribution of a dataset along with key statistical measures		
Scatter Plot	Uses Cartesian coordinates to display values for two variables	<code>DataFrame.plot.scatter()</code> <code>DataFrame.plot(x, y, kind='scatter')</code>	<code>df.plot(x='Height', y='Weight', kind='scatter')</code>

2 Plotting Directly with Matplotlib

Plot Type	Purpose	Matplotlib Function	Example
Line Plot	Shows trends and changes over time	<code>plt.plot()</code>	<code>plt.plot(x, y, color='red', linewidth=2)</code>
Area Plot	Displays data series as filled areas	<code>plt.fill_between()</code>	<code>plt.fill_between(x, y1, y2, color='blue', alpha=0.5)</code>
Histogram	Displays bars representing the data count in each interval/bin	<code>plt.hist()</code>	<code>plt.hist(data, bins=10, color='orange', edgecolor='black')</code>
Bar Chart	Displays data using rectangular bars	<code>plt.bar()</code>	<code>plt.bar(x, height, color='green', width=0.5)</code>
Pie Chart	Displays data as a circular plot divided into slices, representing proportions or percentages of a whole	<code>plt.pie()</code>	<code>plt.pie(sizes, labels=labels, colors=colors, explode=explode)</code>
Box Plot	Displays the distribution of a dataset along with key statistical measures	<code>plt.boxplot()</code>	<code>plt.boxplot(data, notch=True)</code>
Scatter Plot	Uses Cartesian coordinates to display values for two variables	<code>plt.scatter()</code>	<code>plt.scatter(x, y, color='purple', marker='o', s=50)</code>

3 Additional Matplotlib Features

Feature	Purpose	Function	Example
Subplotting	Creating multiple plots on one figure	<code>plt.subplots()</code>	<code>fig, axes = plt.subplots(nrows=2, ncols=2)</code>
Customization	Customizing plots by adding titles, labels, legends, and grids	Various customization functions	<code>plt.title('Title')</code> <code>plt.xlabel('X Label')</code> <code>plt.ylabel('Y Label')</code> <code>plt.legend()</code> <code>plt.grid(True)</code>